Aspen Johnson

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Summary

Data Science student and research assistant with experience in data curation, model validation, technical documentation, and applied research across econometrics, physics, and educational technology. Skilled at reviewing complex information for accuracy, clarity, consistency, and relevance. Interested in AI evaluation, data annotation, prompt testing, and improving AI-assisted tools as a means to get closer to a world where humans and AI can co-exist rather than incite fear.

Education

Valdosta State University

Incoming B.S. in Data Science: Computational Science & Engineering Track

Palm Beach State College

A.A. in STEM: Mechanical & Aerospace Engineering Track — *June 2025 – May 2026, GPA: 3.20*

Publications

An empirical application of improved gradient scaling for score-driven volatility filters

- *Applied Economics Letters*, Taylor & Francis (2025) | Co-authored with **Szabolcs Blazsek** (Mercer University) and **Adam Kobor** (New York University Investment Office).
- Contributed to conceptualization, data curation, formal analysis, methodology, software implementation, validation, visualization, and manuscript preparation for empirical testing of the Hessian-filter scaled Beta-t-EGARCH volatility model across Bitcoin, S&P 500, Gold, U.S. REITs, and CHF/JPY over the period 2010–2025.
- Selected as 1 of 15 research papers to present a lecture at the 2026 Atlanta Research Conference.

Experience

Physics Undergraduate Researcher — Mar 2026 – Present

Valdosta State University — *On-Site/Remote*

- Collaborate with Professor S. Mubin on modeling stock market illiquidity as a bridge toward larger particle simulation research.
- Use Python-based tools including NumPy, Matplotlib, Google Colab, NotebookLM, and Perplexity to support data analysis, visualization, literature review, and research documentation.
- Review model outputs, visualizations, and research notes for accuracy, clarity, and consistency before faculty discussion.

Econometrics Researcher — Jan 2025 – Dec 2025

Mercer University — *On-Site/Remote*

- Sourced and structured financial data from the Bloomberg Terminal, computing return series and descriptive statistics across candidate assets in Excel and R.
- Benchmarked 6+ volatility models (gjrGARCH, sGARCH, Beta-t-EGARCH) in R and GAUSS, selecting optimal asset universe based on data sufficiency and model fit.
- Debugged compilation errors across R and GAUSS environments, documenting terminal logs with line-level annotations for weekly faculty review.
- Validated model outputs and documented errors, inconsistencies, and performance differences across competing volatility models for weekly faculty review.
- Co-authored peer-reviewed manuscript in LaTeX published in *Applied Economics Letters* (Taylor & Francis, 2025).

Financial Analyst Intern — May 2025 – Dec 2025

Zinzino USA — *Remote*

- Reduced chargeback processing time by 87% through workflow digitization and documentation streamlining.
- Refined the chargeback process within the first few weeks and created a clear reference guide to support onboarding, consistency, and cross-functional accuracy.

- Created clear documentation that enabled cross-departmental collaboration and made Finance operations easy for shareholders and upline leaders to understand.
- Reviewed finance-related documentation and case details for completeness, accuracy, and clarity before sharing guidance with internal stakeholders.

Projects

Econometric Modeling in Complex Physical Systems | In Progress

- Investigating the applicability of GARCH-family models to complex physical systems exhibiting variance clustering, including geomagnetic storms, seismic activity, and solar wind turbulence.
- Sourcing and structuring large-scale geophysical time series data from USGS, NOAA, and NCAR databases for empirical volatility modeling.
- Exploring connections between econometric volatility modeling and established physical laws.
- Consulting seismic research labs to learn domain-specific considerations for interpreting earthquake time series and applying volatility-based modeling methods.

DIY Cyberdeck | In Progress

Fusion 360, KiCad, PCB Design, Circuit Design, Electromagnetism, Embedded Systems

- Designing a portable computing device inspired by the uConsole by ClockWorkPi, focusing on modular layout, physical constraints, and technical documentation.
- Learning Fusion 360 to prototype enclosure concepts, component placement, mounting points, and internal spacing.
- Learning KiCad workflows to develop schematic capture, PCB layout, component selection, and hardware design documentation skills.

Leadership, Community Engagement & Extracurriculars

Read2Succeed | *Tutoring young students at the local elementary schools to aid in building literary confidence*

Putnam Mathematics Competition Prep | *Independent Studying*

Division 1 Women's College Basketball Player | *Mercer University (2022 - 2025)*

Division 2 Women's College Basketball Player | *Valdosta State University (Present)*

Youth Sports Volunteer | *Assisted young athletes K-12 in a variety of sports to gain skill and confidence in their respective sp*

Weekly Physics Study Group Member | *Going through **Helliwell's Modern Classical Mechanics** text*

Book Club Fellow | *Selected as one of ten students in Spring 2024 cohort to participate in Mercer University's Economics Dep*

Content Creation | *Created educational livestream content focused on problem sets, coding projects, and technical self-study.*

AWS Campus Prep Series Participant | *Completed a Summer-long series of webinars regarding Amazon Careers and Cloud*

Technical Skills

Programming and Analysis: Python, R, GAUSS, Java, C/C++, JavaScript, Octave, Excel

AI/Data Review: AI response evaluation, prompt testing, data annotation, data labeling, data quality review, pattern recognit

Libraries & Tools: NumPy, Matplotlib, Seaborn, Google Colab, LaTeX, Git, CLI, VS Code

Research & Communication: Literature review, model validation, technical writing, data visualization, educational content d